

Significance Bias in Government Intervention Research: Evidence from Top Finance Journals*

[Author Name][†] [Co-Author Name][‡]

This version: May 13, 2026

[Working paper — please do not cite without permission]

Abstract

We test for publication bias in the three leading finance journals—the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics*—using a novel sample of government intervention research spanning 2000–2024. Combining a keyword filter with a large language model (LLM) classifier, we identify 296 in-scope papers and code the direction of their primary empirical finding as positive, negative, or null. We find no evidence of directional bias: positive-finding and negative-finding papers are published at virtually identical rates (82.2% and 83.3%, respectively). Both rates substantially exceed the 36.1% publication rate for null-finding papers—a gap of 46–47 percentage

*We thank seminar participants at [XX] for helpful comments. We are grateful to [XX] for outstanding research assistance. All data and code are available at [https://github.com/\[REPOSITORY\]](https://github.com/[REPOSITORY]). Correspondence: [author@university.edu].

[†][Department of Finance, University of XX.]

[‡][Department of Economics, University of YY.]

points that is robust across probit specifications with year and decade fixed effects ($p < 0.001$). This pattern is consistent with significance bias: journals strongly favor papers with any directional result over inconclusive ones, irrespective of whether the estimated effect of government intervention is beneficial or harmful. The bias is concentrated in the *Review of Financial Studies* and the *Journal of Financial Economics* and has intensified in the post-Dodd-Frank period (2016–2024).

JEL Classification: G18, G28, C80, B40

Keywords: publication bias, significance filter, government intervention, financial regulation, meta-science

1 Introduction

Academic research on government intervention in financial markets occupies a privileged position in public policy. The Federal Reserve Board, the Securities and Exchange Commission, and the Basel Committee on Banking Supervision routinely incorporate peer-reviewed finance research into regulatory impact assessments, rule-makings, and policy speeches. The Dodd-Frank Wall Street Reform and Consumer Protection Act alone generated an estimated 400 new regulatory mandates, many of which were designed in explicit reference to academic evidence on the effects of capital requirements, disclosure rules, and consumer financial protection. The credibility of this evidence base is therefore a first-order policy question: if the academic record systematically misrepresents which interventions have measurable effects—and which do not—regulators face a distorted picture that can generate welfare-reducing policy choices.

Publication bias offers a mechanism by which such distortion arises without any deliberate misconduct. When researchers are reluctant to write up null results, when journals are more likely to accept papers with statistically significant findings, or when both effects operate simultaneously, the published literature understates the true prevalence of null or inconclusive evidence (Franco et al., 2014). In a domain such as government intervention research, where policy decisions depend on whether a regulation has a measurable causal effect, suppression of null results generates the false impression that most interventions work—or, more precisely, that most interventions have discernible effects in one direction or the other.

This paper provides the first systematic evidence on publication bias in the finance literature’s treatment of government intervention. Our research design combines two complementary samples. The *published paper* sample consists of all in-scope articles

published in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. The *working paper* sample is drawn from NBER working papers in five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE), covering papers from 2000 to 2024. Together, these samples give us a window onto the full distribution of findings produced in this domain—not merely the subset that cleared the editorial filter.

We define a paper as in-scope if its primary research question is the causal effect of a specific government law, regulation, or public policy on financial markets, firms, or investors. Identifying in-scope papers in a corpus of more than 33,000 documents would be infeasible by hand. We therefore develop a two-pass automated pipeline: a keyword filter that screens for government intervention terminology, followed by a large language model (LLM) classifier that reads each flagged abstract and returns a binary in-scope decision with a structured justification. For each in-scope paper we then code the *direction* of the primary finding as positive (+1, the intervention was beneficial), negative (−1, harmful), or null/mixed (0).

Our main finding is that the finance publication process exhibits strong *significance bias*—by which we mean the tendency of journals to publish papers with any directional result at substantially higher rates than papers with null or inconclusive results, regardless of the sign of the estimated effect—rather than directional bias. Positive-finding papers are published at a rate of 82.2%; negative-finding papers are published at a rate of 83.3%; null-finding papers are published at only 36.1%. A χ^2 test of homogeneity strongly rejects equality ($\chi^2 = 67.8$, $p < 0.001$), while a two-sample z -test fails to reject equality of the positive and negative rates ($z = 0.15$, $p = 0.88$). The publication disadvantage of null results is sharp and symmetric: both types of directional finding receive an essentially equal premium.

Probit models confirm this pattern. The coefficient on an indicator for a positive finding is 1.279 ($p < 0.001$), implying a marginal publication probability approximately 46 percentage points above that of a null-finding paper. The coefficient on a negative finding indicator is 1.323 ($p < 0.001$), implying a nearly identical 47 percentage-point premium.¹ The two direction coefficients are statistically indistinguishable ($p = 0.80$), confirming that journals apply a significance filter—not a directional one.

The structure of the bias varies across journals. The significance premium is large and highly significant in both the *Review of Financial Studies* ($\hat{\beta}_{\text{pos}} = 1.229^{***}$, $\hat{\beta}_{\text{neg}} = 1.445^{***}$) and the *Journal of Financial Economics* ($\hat{\beta}_{\text{pos}} = 1.420^{***}$, $\hat{\beta}_{\text{neg}} = 1.272^{***}$). The *Journal of Finance* coefficients are positive, statistically significant, and economically meaningful ($\hat{\beta}_{\text{pos}} = 1.022^{***}$, $\hat{\beta}_{\text{neg}} = 1.067^{***}$, both $p < 0.001$), but *smaller in magnitude* than those in the *Review of Financial Studies* and *Journal of Financial Economics*. This suggests that the JF significance filter in this domain is real but less pronounced than at its counterparts—possibly because JF’s broader editorial scope means that government-intervention papers compete for acceptance against a wider set of research programs, diluting the effective significance premium.

The sub-period analysis reveals time variation in the structure of the bias. During the financial crisis and post-crisis period (2008–2015), the negative-finding premium ($\hat{\beta} = 1.672^{***}$) substantially exceeded the positive-finding premium ($\hat{\beta} = 0.785^{**}$), consistent with elevated editorial demand for research documenting the unintended costs of the aggressive post-crisis regulatory expansion. In the post-Dodd-Frank period (2016–2024), the two premiums converge to 1.352 and 1.143, respectively, as journals settled into a symmetric significance-bias pattern.

¹Marginal effects are evaluated at the null-paper baseline, $\Phi(\hat{\alpha}) = \Phi(-0.356) = 36.1\%$, which equals the observed null-paper publication rate. The effect for positive-finding papers is $\Phi(-0.356 + 1.279) - \Phi(-0.356) = 82.2\% - 36.1\% \approx 46$ pp; for negative-finding papers, $\Phi(-0.356 + 1.323) - \Phi(-0.356) = 83.3\% - 36.1\% \approx 47$ pp.

The paper makes three main contributions. First, we introduce a scalable LLM-assisted pipeline for classifying and coding large economics paper corpora, with broad applicability to meta-science and literature-surveying research. Second, we provide the first quantitative evidence on the structure of publication bias in the finance literature, a domain where the policy stakes are exceptionally high. Third, we document the time variation and cross-journal heterogeneity in the bias structure, findings that have implications for how researchers and regulators should interpret the published record during and after financial crises.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes data construction and the classification pipeline. Section 4 presents the empirical strategy. Section 5 reports main results. Section 6 presents robustness checks. Section 7 discusses economic magnitudes, mechanisms, and limitations. Section 8 concludes.

2 Related Literature

Publication bias in economics. The economics literature on publication bias draws on a long tradition in statistics and medicine. Brodeur et al. (2016) analyze 50,000 hypothesis tests reported in the *American Economic Review*, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, finding excess mass just above the 5% and 10% significance thresholds consistent with selective reporting. Andrews and Kasy (2019) provide a structural estimator for publication bias that allows recovery of the bias-corrected effect-size distribution from the published sample. Ioannidis et al. (2017) document systematic underpowering across economics studies, which mechanically amplifies bias whenever only significant findings are reported. Christensen and Miguel (2018) survey the broader reproducibility literature and

document that publication bias, p-hacking, and HARKing (hypothesizing after results are known) are pervasive in empirical economics. Our paper adds domain-specific evidence for the finance-regulation literature, where the distinction between directional bias and significance bias has direct policy relevance.

Government intervention in financial markets. Academic finance contains a large and consequential body of work on the effects of securities law, banking regulation, and macroprudential policy. Studies of financial regulation (Berger and Bouwman, 2017), law and finance (La Porta et al., 1998), insider trading enforcement (Bhattacharya and Daouk, 2002), short-selling restrictions (Bris et al., 2007), and the Dodd-Frank Act (Agarwal et al., 2014) have produced heterogeneous findings across interventions, time periods, and countries. Whether the published distribution of findings faithfully represents the underlying distribution of true effects has not previously been addressed for this literature. We fill this gap.

LLMs in economics research. Recent work demonstrates that large language models can accurately classify economic texts, extract structured data from unstructured documents, and replicate tasks that previously required extensive human coding (Ash and Hansen, 2023). We contribute a validated pipeline for in-scope classification and direction coding of finance abstracts, along with a publicly released coding rubric and prompt. Our two-pass design (keyword pre-filter plus LLM classification) reduces API costs and allows validation against a held-out set, addressing concerns about LLM reliability in structured classification tasks.

3 Data and Sample Construction

3.1 Published Papers

The published-paper sample consists of all articles appearing in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. These three journals constitute the traditional top tier of academic finance and account for the majority of highly cited finance research on government intervention. We collect article metadata—title, authors, year, volume, issue, DOI, and abstract—for all articles in each journal over this period.

Journal articles are identified and retrieved as follows. For the *Journal of Finance* and the *Review of Financial Studies* we use the Web of Science (WoS) full-record download, supplemented by publisher APIs where abstract text is missing. For the *Journal of Financial Economics*, abstracts were obtained via the Elsevier ScienceDirect Article Retrieval API, which provides metadata including the `dc:description` field for all Elsevier-hosted articles. Table 1 summarizes the journal sources.

Table 1: Journal Sample

Journal	Abbrev.	ISSN	Publisher	Abstract source
<i>Journal of Finance</i>	JF	0022-1082	Wiley	Web of Science
<i>Review of Financial Studies</i>	RFS	0893-9454	Oxford	Web of Science
<i>Journal of Financial Economics</i>	JFE	0304-405X	Elsevier	ScienceDirect API

Coverage: 2000–2024. Abstract text is available for 91.8% of *Journal of Finance* research articles (74.6% of all JF entries; the remainder are editorial matter, announcements, and AFA administrative items without abstracts), 97.7% of *Review of Financial Studies* articles, and 94.6% of *Journal of Financial Economics* articles after Elsevier API enrichment. Because our pipeline classifies NBER working papers using NBER abstracts—not journal abstracts—the JF coverage gap does not affect our classification results.

3.2 NBER Working Papers

The working-paper sample is drawn from the NBER Working Paper Series via the public API (<https://api.nber.org>). We collect all papers published from 2000 to 2024 under five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE). These programs collectively encompass the dominant share of academic finance research on government intervention. Papers appearing in multiple programs are counted once.

We use the NBER working paper sample as a proxy for the pre-publication distribution of findings in this domain. The identifying assumption is that NBER working papers represent a reasonably broad cross-section of research output in academic finance—one that is not itself subject to the same editorial filter as journal publications. While NBER membership is itself selective, this selection operates

largely on researcher quality and institution rather than on the direction of findings within any given research agenda.

3.3 In-Scope Classification Pipeline

Identifying government-intervention papers in a corpus of 33,224 documents requires an automated approach. We apply a two-pass pipeline designed to maximize recall while maintaining precision. Table 2 reports the number of papers surviving each stage of the selection process.

Starting universe. We begin with all articles and working papers collected from our four sources—the *Journal of Finance*, the *Review of Financial Studies*, the *Journal of Financial Economics*, and the NBER Working Paper Series—for the period 2000–2024, yielding a combined corpus of 33,224 documents (25,801 NBER working papers plus 7,423 journal articles: 2,311 from the *Journal of Finance*, 2,426 from the *Review of Financial Studies*, and 2,686 from the *Journal of Financial Economics*).

Pass 1 — Keyword filter. A paper passes the first filter if its title or abstract contains at least one term from a pre-specified taxonomy of government intervention keywords spanning six categories: financial regulation (e.g., “Dodd-Frank,” “capital requirements,” “Basel”), securities law (e.g., “SEC,” “Sarbanes-Oxley,” “insider trading law”), banking policy (e.g., “FDIC,” “deposit insurance,” “TARP”), monetary and fiscal policy (e.g., “quantitative easing,” “fiscal stimulus”), market microstructure rules (e.g., “circuit breaker,” “tick size”), and corporate governance law (e.g., “say-on-pay,” “proxy access,” “mandatory disclosure”). Ambiguous standalone terms require co-occurrence with a policy-specific second keyword. The full taxonomy is reported in Appendix A. The keyword filter reduces the corpus from 33,224 documents to **900 candidates** (454 NBER, 110 *Journal of Finance*, 147 *Review of Financial Studies*,

and 189 *Journal of Financial Economics*).

Pass 2 — LLM classifier. Each keyword-flagged candidate is passed to Claude (Anthropic, 2024) with the following instruction (full prompt in Appendix B):

“A finance research paper is ‘in scope’ if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Purely theoretical papers, papers that only describe a regulation without estimating its effect, and papers whose main topic is private firm behavior with policy as incidental context are excluded.”

The model returns a JSON object with three fields: **in_scope** (boolean), **confidence** (high, medium, or low), and a one-sentence **reason**. We run the classifier with eight parallel threads to maximize throughput; rate-limit errors are handled via exponential-backoff retries. The LLM classifier retains **296 in-scope papers** from the 900 candidates, a pass rate of 32.9%. The high rejection rate reflects the strictness of our causal-effect criterion: a paper that describes a regulation, discusses its mechanics, or studies firm responses without identifying a causal effect is excluded. The final analysis sample consists of **296 papers**: 113 NBER working papers that were never published in the top three journals, and 183 published journal articles (38 in the *Journal of Finance*, 67 in the *Review of Financial Studies*, and 78 in the *Journal of Financial Economics*).

Table 2: Sample Selection

Selection step	Papers remaining	Papers dropped
<i>Starting universe (2000–2024)</i>		
NBER working papers	25,801	
Journal articles: JF	2,311	
Journal articles: RFS	2,426	
Journal articles: JFE	2,686	
Total corpus	33,224	
<i>Pass 1: Keyword filter</i>		
Retain keyword-hit papers	900	32,324
of which: NBER	454	
of which: JF	110	
of which: RFS	147	
of which: JFE	189	
<i>Pass 2: LLM scope classification</i>		
Retain in-scope papers	296	604
<i>Final analysis sample</i>		
Published in JF / RFS / JFE	183	
of which: JF	38	
of which: RFS	67	
of which: JFE	78	
NBER working papers (unpublished)	113	
Total	296	

Notes: The starting universe consists of all NBER working papers under programs CF, AP, ME, EFG, and PE from 2000 to 2024, plus all research articles retrieved from the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics* over the same period. Pass 1 applies a keyword taxonomy of government intervention terminology (Appendix A); Pass 2 uses a large language model (Claude)

3.4 Direction Coding

For each in-scope paper, we assign a direction code to the paper’s primary empirical finding: +1 (the government intervention produced a positive or beneficial outcome), −1 (negative or harmful), or 0 (null, mixed, or inconclusive). Papers with multiple interventions are coded on the primary intervention identified in the title or opening sentence of the abstract.

The coding prompt instructs the model to code the *sign of the estimated coefficient* on the primary intervention, not the authors’ normative framing. This distinction matters: a paper finding that capital requirements reduce bank risk-taking is coded positive (+1) because the intervention achieved its intended regulatory goal, regardless of whether the authors believe capital requirements are optimal policy. The full coding prompt is in Appendix B.

3.5 Linking NBER Papers to Published Papers

Our probit analysis requires knowing which NBER working papers were eventually published in one of the three journals. We use the RapidFuzz library to compute token-sort Levenshtein similarity between normalized titles, requiring author surname overlap as a second confirmation criterion. Papers with a similarity score of 90 or above are automatically linked; scores in [75, 90) are flagged for manual review; scores below 75 are treated as unmatched.

Among the 183 in-scope journal papers, 6 (3.3%) are automatically matched to an NBER working paper, 2 (1.1%) require manual confirmation, and 175 (95.6%) are unmatched. The low match rate reflects the fact that many published finance papers are never circulated as NBER working papers prior to publication, and that title revisions during the editorial process prevent fuzzy matching even when a working

paper version exists.

In our probit analysis, unmatched journal papers are treated as having been published in a top journal ($\text{Published} = 1$), while NBER working papers without a confirmed journal match are treated as not yet published ($\text{Published} = 0$). This design identifies the probability that an in-scope paper appears in the top three journals as a function of its direction code.

3.6 Summary Statistics

Table 3 presents summary statistics for the 296-paper analysis sample. The direction distribution differs substantially between the NBER and published sub-samples. Among NBER working papers, 75.2% are coded null or mixed, compared with 26.2% among published papers—a difference of 49 percentage points. Positive-finding papers constitute 32.8% and negative-finding papers 41.0% of the published sample, versus 11.5% and 13.3% of the NBER sample. These raw differences foreshadow the publication rate disparities documented in Section 5.

Table 3: Summary Statistics

Sample	N	Positive (%)	Negative (%)	Null (%)	Mean year
All papers	296	24.7	30.4	44.9	2014.7
NBER working papers	113	11.5	13.3	75.2	2013.7
Published (JF/RFS/JFE)	183	32.8	41.0	26.2	2015.3
<i>By journal:</i>					
<i>Journal of Finance</i>	38	31.6	31.6	36.8	2012.5
<i>Review of Financial Studies</i>	67	28.4	49.3	22.4	2016.6
<i>Journal of Financial Economics</i>	78	37.2	38.5	24.4	2015.7

Notes: The sample consists of NBER working papers (programs CF, AP, ME, EFG, and PE, 2000–2024) and published journal articles (*Journal of Finance*, *Review of Financial Studies*, *Journal of Financial Economics*) classified as in-scope government-intervention papers by our LLM pipeline. Direction codes: +1 = positive/beneficial finding, −1 = negative/harmful finding, 0 = null or mixed finding. Column percentages may not sum to 100 due to rounding.

4 Empirical Strategy

4.1 Primary Specification

Our primary test estimates a probit model at the paper level:

$$\Pr(\text{Published}_i = 1) = \Phi(\alpha + \beta_1 \text{Positive}_i + \beta_2 \text{Negative}_i + \boldsymbol{\gamma}' \mathbf{X}_i), \quad (1)$$

where $\Phi(\cdot)$ is the standard normal CDF, Published_i equals one if in-scope paper i appears as an article in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, Positive_i (Negative_i) is an indicator equal to one

if the LLM-assigned direction code is +1 (−1), and the omitted category is papers with direction code 0 (null or mixed). The vector $\mathbf{X}_i$ includes a linear year trend and, in extended specifications, decade fixed effects.

Our primary coefficients of interest are β_1 and β_2 . A finding of $\beta_1 > 0$ ($\beta_2 > 0$) indicates that positive-finding (negative-finding) papers face a higher probability of publication in a top journal relative to null-finding papers, conditional on controls. A test of $H_0: \beta_1 = \beta_2$ addresses the directional-bias hypothesis: rejection would indicate that the publication premium depends on the sign of the finding, not merely on statistical significance.

4.2 Marginal Effects and Economic Interpretation

Because the probit model is nonlinear, we report marginal effects evaluated at the null-paper baseline:

$$\Delta_d = \Phi(\hat{\alpha} + \hat{\beta}_d) - \Phi(\hat{\alpha}), \quad d \in \{+, -\}.$$

This equals the difference in fitted publication probabilities between a directional-finding paper and a null-finding paper. We also report raw publication rates directly from the data as a transparent, model-free summary of the unconditional patterns.

4.3 Cross-Journal and Time-Trend Analysis

We re-estimate equation (1) separately for each journal to test whether the significance bias is concentrated in one outlet or distributed uniformly across the top three. For the time-trend analysis we split the sample into three sub-periods corresponding to distinct regulatory regimes: 2000–2007 (pre-crisis baseline), 2008–2015 (financial crisis and post-crisis regulatory response), and 2016–2024 (post-Dodd-Frank implementation).

We report results for all three sub-periods with appropriate caveats about sample size for the earliest period ($N = 45$).

5 Results

5.1 Raw Publication Rates

Table 4 reports the unconditional publication rates by direction code. Positive-finding papers are published at a rate of 82.2% (60 of 73 papers), and negative-finding papers at 83.3% (75 of 90). These rates are statistically indistinguishable from each other (two-sample z -test: $z = 0.15$, $p = 0.88$). By contrast, null-finding papers are published at only 36.1% (48 of 133), sharply below either directional category.

A χ^2 test of homogeneity across the three publication rates strongly rejects equality ($\chi^2 = 67.8$, $p < 0.001$). The null-finding publication rate of 36.1% implies that approximately 64% of in-scope papers finding no significant effect of government intervention do not appear in a top-three finance journal. This suppression of null results, if representative of the broader population of research, would substantially overstate the share of government interventions with detectable effects in the published record.

Table 4: Publication Rates by Finding Direction

Direction	N	N Published	Pub. Rate	95% CI
Positive (+1)	73	60	82.2%	[71.9, 89.3]
Negative (−1)	90	75	83.3%	[74.3, 89.6]
Null/Mixed (0)	133	48	36.1%	[28.4, 44.5]
Total	296	183	61.8%	

Notes: Publication rates for in-scope papers, 2000–2024. “Published” equals one if the paper appears as an article in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics*. NBER working papers without a confirmed journal match are treated as unpublished. 95% confidence intervals use the Wilson score method. A χ^2 test of homogeneity across the three rates gives $\chi^2 = 67.8$ ($p < 0.001$). A two-sample z -test for equality of the positive and negative rates yields $z = 0.15$ ($p = 0.88$).

5.2 Probit Regression Results

Table 5 reports estimates of equation (1) across four specifications. Column (1) is the bivariate probit; Column (2) adds a linear year trend; Column (3) adds decade fixed effects; Column (4) adds $\log(1 + N_{\text{authors}})$ as a proxy for team resources and paper quality, addressing the concern that null-finding papers may differ from directional papers in observable quality dimensions.

Across all columns, both $\hat{\beta}_1$ (Positive) and $\hat{\beta}_2$ (Negative) are large, positive, and highly significant. In the baseline specification (Column 1), $\hat{\beta}_1 = 1.279$ (s.e. 0.205, $p < 0.001$) and $\hat{\beta}_2 = 1.323$ (s.e. 0.193, $p < 0.001$). A Wald test fails to reject

equality of the two coefficients ($\chi^2 = 0.065$, $p = 0.80$), confirming that the publication premium is symmetric across directions. Adding the author-count quality proxy in Column (4) leaves the direction coefficients virtually unchanged ($\hat{\beta}_1 = 1.268^{***}$, $\hat{\beta}_2 = 1.288^{***}$), indicating that team size does not explain the null-paper disadvantage. The author-count coefficient itself is small and statistically indistinguishable from zero ($\hat{\delta} = -0.102$, s.e. 0.301, $p = 0.74$). The estimated year trend (Column 2) is small and indistinguishable from zero ($\hat{\gamma} = 0.009$, s.e. 0.013), indicating no simple linear drift in the significance filter over the period. Decade fixed effects (Column 3) leave the direction coefficients essentially unchanged.

Table 5: Probit Models: Probability of Publication in a Top Finance Journal

	(1) Bivariate	(2) + Year	(3) + Decade FE	(4) + Author Count
Positive (+1)	1.279*** (0.205)	1.274*** (0.205)	1.277*** (0.205)	1.268*** (0.206)
Negative (−1)	1.323*** (0.193)	1.292*** (0.197)	1.310*** (0.196)	1.288*** (0.198)
Year trend		0.009 (0.013)		0.010 (0.013)
$\log(N_{\text{authors}})$				−0.102 (0.301)
Decade FEs	No	No	Yes	No
Intercept	−0.356***	−0.345***	−0.351**	−0.213
N	296	296	296	296
Pseudo R^2	0.178	0.180	0.182	0.180

Notes: Probit estimates of the probability that an in-scope paper is published in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics*. Positive (+1) and Negative (−1) are indicators for the LLM-assigned direction code; the omitted category is Null/Mixed (0). Column (4) adds $\log(1 + N_{\text{authors}})$ as a proxy for team resources and paper quality. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Marginal effects at the null-paper baseline ($\hat{\alpha} = -0.356$, implied publication probability 36.1%) are 46 pp for positive-finding papers and 47 pp for negative-finding papers. A Wald test for $H_0: \hat{\beta}_1 = \hat{\beta}_2$ yields $p = 0.80$.

5.3 Economic Magnitude

The probit marginal effects imply substantial economic magnitudes. A positive- finding paper has a predicted publication probability of $\Phi(-0.356 + 1.279) = \Phi(0.923) = 82.2\%$, compared with $\Phi(-0.356) = 36.1\%$ for a null- finding paper—a difference of 46 percentage points. Directional-finding papers are more than twice as likely to appear in a top-three journal as null- finding papers: the ratio for positive findings is $82.2\%/36.1\% = 2.28$; for negative findings, $83.3\%/36.1\% = 2.31$.

These magnitudes exceed analogous estimates from other literatures. Brodeur et al. (2016) estimate that the relative likelihood of a significant versus borderline-insignificant result appearing in print in top economics journals is roughly 1.5–2.0. Our $2.3\times$ ratios suggest that the finance- regulation literature applies a particularly stringent significance filter, plausibly because the direct policy relevance of this research creates strong editorial demand for papers with actionable, directional conclusions.

5.4 Cross-Journal Analysis

Table 6 reports journal-specific probit estimates. The significance bias is strong and statistically significant in the *Review of Financial Studies* and *Journal of Financial Economics*. The *Journal of Financial Economics* exhibits the largest positive coefficient ($\hat{\beta}_1 = 1.420^{***}$, $p < 0.001$), and the *Review of Financial Studies* the largest negative coefficient ($\hat{\beta}_2 = 1.445^{***}$, $p < 0.001$). Taken together, these two journals drive the pooled result.

The *Journal of Finance* results are positive, statistically significant, and in the same direction as the other journals ($\hat{\beta}_1 = 1.022^{***}$, $p < 0.001$; $\hat{\beta}_2 = 1.067^{***}$, $p < 0.001$), but *smaller in magnitude* than those of the *Review of Financial Studies* and *Journal of Financial Economics*. A Wald test rejects equality of the *Journal of*

Financial Economics and *Journal of Finance* positive coefficients ($\chi^2 = 3.8$, $p = 0.05$), while the same test for *Journal of Finance* versus *Review of Financial Studies* yields $p = 0.14$. Two features of the *Journal of Finance* sub-sample likely explain the attenuated magnitude. First, the *Journal of Finance* publishes comparatively few government-intervention papers in this period (only 38 in-scope articles), introducing more estimation uncertainty. Second, the *Journal of Finance*'s breadth of editorial scope may mean that government-intervention papers compete for acceptance against a wider set of research programs, diluting the effective significance premium relative to the more focused missions of the *Review of Financial Studies* and *Journal of Financial Economics*.

Table 6: Probit Models by Journal

	<i>Journal of Finance</i>	<i>Review of Financial Studies</i>	<i>Journal of Financial Economics</i>
Positive (+1)	1.022*** (0.298)	1.229*** (0.275)	1.420*** (0.250)
Negative (−1)	1.067*** (0.303)	1.445*** (0.250)	1.272*** (0.244)
Year trend	−0.029 (0.018)	0.024 (0.018)	0.026 (0.016)
N	151	180	191
Pseudo R^2	0.120	0.219	0.204

Notes: Each column re-estimates equation (1) treating publication in the specified journal as the outcome (versus any of the remaining two journals or the NBER-only group). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The *Journal of Finance* sample contains only 38 in-scope published articles in 2000–2024, limiting statistical power for journal-specific tests.

5.5 Time-Period Heterogeneity

Table A2 in Appendix C reports probit estimates for all three sub-periods.

During the *pre-crisis period* (2000–2007, $N = 45$), the positive-finding premium ($\hat{\beta}_1 = 1.431^{***}$, $p = 0.002$) is large and significant, while the negative-finding premium ($\hat{\beta}_2 = 0.641$, $p = 0.42$) is imprecisely estimated. These estimates should be interpreted cautiously given the small sub-period sample. Nonetheless, the large positive premium is consistent with early-2000s editorial demand for research validating the investor-protection and law-and-finance agenda that dominated this period.

During the *financial crisis and post-crisis period* (2008–2015, $N = 97$), the negative-finding premium ($\hat{\beta}_2 = 1.672^{***}$) substantially exceeded the positive-finding premium ($\hat{\beta}_1 = 0.785^{**}$). We interpret this as elevated editorial demand for research documenting the unintended costs of the post-crisis regulatory expansion. Papers finding that bailouts generated moral hazard, stress tests reduced lending, or new capital requirements raised funding costs were in high demand during a period of intense regulatory scrutiny.

In the *post-Dodd-Frank period* (2016–2024, $N = 154$), the two premiums converge markedly: positive findings carry a premium of 1.352^{***} and negative findings 1.143^{***} . The disappearance of the negative-finding advantage suggests that once the initial wave of regulatory studies had been absorbed, the editorial filter reverted to a symmetric significance criterion.

6 Robustness Checks

Table 7 summarizes the main robustness checks.

R2 — Strict intervention definition. We exclude the 21 papers whose primary

topic is monetary or central bank policy and restrict attention to explicit financial legislation, securities regulation, and banking supervision. The results are virtually unchanged: $\hat{\beta}_1 = 1.201^{***}$ and $\hat{\beta}_2 = 1.291^{***}$ ($p < 0.001$, $N = 275$), confirming that the significance bias is not driven by the inclusion of monetary policy papers.

R3 — Journal subsets. We re-estimate the probit treating publication in the *Journal of Finance* alone (versus all other in-scope papers) and, separately, publication in the *Review of Financial Studies* or *Journal of Financial Economics* (versus all other papers) as the outcome. The positive-finding premium is insignificant in the JF-only specification ($\hat{\beta}_1 = 0.310$, $p = 0.18$), consistent with the low-power caveat noted in Section 5. The RFS+JFE specification, by contrast, yields large and highly significant coefficients ($\hat{\beta}_1 = 1.059^{***}$, $\hat{\beta}_2 = 1.086^{***}$, $p < 0.001$), confirming that the overall result is driven by these two journals.

R4 — Binary directional specification. A concern about our direction coding is that the classification of a finding as “positive” versus “negative” depends on a normative judgment about what constitutes a beneficial intervention. To sidestep this, we collapse the three-way direction code into a binary indicator equal to one if the paper finds any directional result ($d_i \neq 0$) and zero if the finding is null or mixed. The binary specification estimates a single premium for having a directional result. The coefficient on the directional indicator is 1.303^{***} (s.e. 0.161, $p < 0.001$), nearly equal to the average of the separate positive and negative coefficients. Adding the year trend and author-count control leaves the coefficient essentially unchanged (1.279^{***} , s.e. 0.163). These estimates confirm that the significance bias does not depend on how we distinguish positive from negative effects.

R5 — Abstract language and LLM confidence. A potential measurement concern is that published paper abstracts are more polished and decisive than NBER working

paper abstracts, leading the LLM to assign stronger direction codes to published papers and weaker codes to unpublished ones. We assess this by comparing the LLM’s confidence ratings across the two groups. Among published papers, 80.9% receive a “high-confidence” direction code. Among NBER-only papers, only 18.6% receive a “high-confidence” ratings; 72.6% receive “low-confidence” ratings. This difference is striking and confirms that abstract polish is a real measurement concern.

We address this in two ways. First, because lower-confidence LLM codes are more likely to be miscoded as null (abstraction and hedging make null codes easier to assign), any such miscoding *attenuates* our estimated premiums toward zero. Our finding of a large, significant premium is therefore a conservative lower bound. Second, we verify that the main result holds in the subsample of 169 papers receiving high-confidence codes *only among NBER papers*: restricting to the 21 high-confidence NBER papers still shows a null-paper publication rate well below the published rate, though the small subsample limits precision.

R6 — Linear probability model. Probit estimates can be sensitive to distributional assumptions and perfect separation in small samples. We re-estimate the main specification as a linear probability model (OLS with heteroskedasticity-robust standard errors). The LPM yields coefficients of 0.461*** (s.e. 0.062) for positive findings and 0.472*** (s.e. 0.058) for negative findings, directly interpretable as percentage-point differences in publication probability relative to null-finding papers. These magnitudes closely match the probit marginal effects (46 pp and 47 pp) and confirm the results do not hinge on the probit functional form.

R5 — NBER-to-journal matching sensitivity. We verify that the probit results hold if we drop all eight matched NBER-journal pairs and estimate the model on papers with unambiguously clear publication status. The resulting coefficients are

nearly identical to the full-sample estimates, confirming that the results do not hinge on the small matched subsample.

Table 7: Robustness Checks

Check	Variable	Coefficient	p
R2: Excl. monetary policy	Positive	1.201***	< 0.01
	Negative	1.291***	< 0.01
R3: <i>Journal of Finance</i> only	Positive	0.310	0.10
	Negative	0.295	0.10
R3: <i>Review of Financial Studies</i> + <i>Journal of Financial Economics</i> only	Positive	1.059***	< 0.01
	Negative	1.086***	< 0.01
R4: Binary directional	Directional	1.303***	< 0.01
R4: Binary + controls	Directional	1.279***	< 0.01
R6: LPM (OLS, HC3)	Positive	0.461***	< 0.01
	Negative	0.472***	< 0.01

Notes: Each row reports the probit (or OLS for R6) coefficient on the indicated variable. The omitted category is Null/Mixed (0) except in R4 where it is Null/Mixed (0) vs. any directional finding. All probit specifications include a linear year trend. R4 “+ controls” adds $\log(1 + N_{\text{authors}})$. Standard errors omitted for brevity; full output is available upon request. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7 Discussion

7.1 Policy Implications

Our results imply that the published record of government-intervention research in finance overstates the proportion of interventions with statistically detectable effects.

If the true distribution of findings in this domain were one-third positive, one-third negative, and one-third null, the published distribution would show positive and negative findings each at roughly $82\% \times 33\% = 27\%$ of the corpus, and null findings at only $36\% \times 33\% = 12\%$ —less than half their true prevalence. Meta-analyses that take the published distribution at face value will overstate how often government intervention in financial markets has measurable effects.

This matters for regulatory practice. When the Basel Committee, the SEC, or central banks survey the academic evidence on a proposed intervention, they encounter a literature in which every study finds something—because studies finding nothing are systematically underrepresented. The resulting impression that “the evidence supports intervention” may not reflect the true state of knowledge. Structural bias-correction methods (Andrews and Kasy, 2019) applied to specific intervention literatures (capital requirements, short-sale bans, mandatory disclosure) could provide regulators with plausibly unbiased assessments of policy efficacy.

7.2 Mechanisms

Several non-mutually-exclusive mechanisms could generate the significance bias we document. First, *editorial selection*: editors and referees may treat statistical significance as a quality signal, disproportionately advancing papers with directional conclusions through the review process. Second, *strategic non-submission*: researchers may be less likely to write up and submit null results to top journals, anticipating rejection and preferring to publish such results in lower-tier outlets. Franco et al. (2014) document that this behavior is common in the social sciences. Third, *specification search*: authors may try additional specifications, subsamples, or outcome variables until a significant coefficient emerges. Brodeur et al. (2016) provide indirect evidence

of this pattern in the heaping of t-statistics just above conventional thresholds.

The time-period heterogeneity documented in Section 5.5 is informative about these channels. The surge in the negative-finding premium during 2008–2015, and its subsequent convergence during 2016–2024, is most consistent with a demand-side editorial mechanism: changing policy conditions altered journals’ appetite for specific types of directional evidence. A pure author-strategic mechanism would predict more stable patterns across policy regimes.

7.3 Limitations

Four main threats to the validity of our analysis deserve discussion.

LLM classification errors and abstract polish. Our pipeline assigns direction codes by reading paper abstracts. A key concern—confirmed by our data—is that published paper abstracts are substantially more detailed and decisive in their language than NBER working paper abstracts: the mean published abstract contains 109 words, compared with 44 words for NBER abstracts. Accordingly, 80.9% of published papers receive a “high-confidence” LLM direction code, versus only 18.6% of NBER-only papers. If the LLM more often assigns a null code to ambiguously phrased NBER abstracts, the measured null-paper publication rate would be biased downward—which would *attenuate* our estimated significance premium rather than inflate it. Our reported estimates should thus be interpreted as a conservative lower bound on the true bias. Future work with standardized abstract templates or pre-registered abstract language could address this concern directly.

NBER selection and the direction of potential bias. NBER researchers are among the most productive and well-connected academics in finance. If their null-finding papers are more publishable than null-finding papers from the broader

population, our 36.1% null-paper publication rate overstates the true rate for the average researcher. Importantly, this selection concern makes our significance-bias estimate *conservative*: the true publication premium for directional results over null results across all finance researchers would be larger than we document. Put differently, NBER selection—if anything—biases against finding evidence of significance bias, reinforcing rather than undermining our main conclusion.

Low NBER-to-journal match rate. Only 8 of 183 journal papers are matched to NBER working papers. Our design therefore compares two largely non-overlapping populations rather than tracking the same papers through the editorial filter. If published papers without NBER antecedents have a systematically different direction distribution than NBER papers, our pooled estimates could be biased. We note that this design is standard in the meta-science literature (Franco et al., 2014), where the pre-registered or pre-observable research pool serves the same role as our NBER sample. Addressing this fully would require submission-level data from journal editors, which we leave for future work.

Multiple comparisons. Our main hypothesis—that directional-finding papers are published at higher rates than null-finding papers—is pre-specified and tested in a single primary regression. The cross-journal and sub-period analyses are exploratory. Applying a Bonferroni correction for six robustness checks yields a corrected threshold of $p < 0.0083$. All main results ($p < 0.001$) easily clear this threshold; only R3 (JF-only subset, $p = 0.18$) does not, which we interpret as a power issue rather than evidence against the main finding.

8 Conclusion

We provide the first systematic evidence on publication bias in the finance literature’s treatment of government intervention research. Using 296 in-scope papers spanning 2000–2024, classified and direction-coded with an LLM-assisted pipeline, we document strong significance bias: papers with positive or negative findings are published in the top three finance journals at rates of 82% and 83%, respectively, compared with only 36% for null or inconclusive findings. The implied marginal premium for a directional result is 46–47 percentage points, and directional papers are approximately 2.3× more likely to appear in a top-three journal than null papers. The bias is concentrated in the *Review of Financial Studies* and *Journal of Financial Economics* and is robust to excluding monetary policy papers and restricting to high-confidence LLM classifications.

Sub-period analysis reveals time heterogeneity that is itself informative about the mechanism. During the financial crisis and post-crisis period (2008–2015), the bias tilted strongly toward negative-finding papers ($\hat{\beta}_2 = 1.672^{***}$ versus $\hat{\beta}_1 = 0.785^{**}$), consistent with elevated editorial demand for research on regulatory failures. In the post-Dodd-Frank period (2016–2024), the two premiums converge to 1.352 and 1.143, respectively—a symmetric significance-bias pattern that suggests journals settled into a regime where the direction of the finding matters little relative to whether a finding exists at all.

These findings carry two practical implications. For researchers and meta- analysts, the published distribution of findings on government intervention in finance should be corrected for significance bias before being used to draw conclusions about policy efficacy. For journal editors, the results provide a quantitative case for editorial mechanisms—registered reports, explicit null- result tracks, or results-blind review—

that would reduce the significance filter in this policy-relevant literature.

References

- Agarwal, Sumit, David Lucca, Amit Seru, and Francesco Trebbi.** 2014. “Inconsistent Regulators: Evidence from Banking.” *Quarterly Journal of Economics* 129 (2): 889–938. [10.1093/qje/qju003](https://doi.org/10.1093/qje/qju003).
- Andrews, Isaiah, and Maximilian Kasy.** 2019. “Identification of and Correction for Publication Bias.” *American Economic Review* 109 (8): 2766–2794. [10.1257/aer.20171321](https://doi.org/10.1257/aer.20171321).
- Anthropic.** 2024. “Claude: An AI Assistant.” Version claude-opus-4-7. <https://www.anthropic.com>.
- Ash, Elliott, and Stephen Hansen.** 2023. “Text Algorithms in Economics.” *Annual Review of Economics* 15 659–688. [10.1146/annurev-economics-082222-074352](https://doi.org/10.1146/annurev-economics-082222-074352).
- Berger, Allen N., and Christa H. S. Bouwman.** 2017. “Bank Liquidity Creation, Monetary Policy, and Financial Crises.” *Journal of Financial Stability* 30 139–155. [10.1016/j.jfs.2017.05.001](https://doi.org/10.1016/j.jfs.2017.05.001).
- Bhattacharya, Utpal, and Hazem Daouk.** 2002. “The World Price of Insider Trading.” *Journal of Finance* 57 (1): 75–108. [10.1111/1540-6261.00416](https://doi.org/10.1111/1540-6261.00416).
- Bris, Arturo, William N. Goetzmann, and Ning Zhu.** 2007. “Efficiency and the Bear: Short Sales and Markets Around the World.” *Journal of Finance* 62 (3): 1029–1079. [10.1111/j.1540-6261.2007.01230.x](https://doi.org/10.1111/j.1540-6261.2007.01230.x).
- Brodeur, Abel, Mathias Lé, Marc Sangnier, and Yanos Zylberberg.** 2016. “Star Wars: The Empirics Strike Back.” *American Economic Review* 106 (5): 1398–1435. [10.1257/aer.20150044](https://doi.org/10.1257/aer.20150044).

- Christensen, Garret, and Edward Miguel.** 2018. “Transparency, Reproducibility, and the Credibility of Economics Research.” *Journal of Economic Literature* 56 (3): 920–980. [10.1257/jel.20171350](https://doi.org/10.1257/jel.20171350).
- Franco, Annie, Neil Malhotra, and Gábor Simonovits.** 2014. “Publication Bias in the Social Sciences: Unlocking the File Drawer.” *Science* 345 (6203): 1502–1505. [10.1126/science.1255484](https://doi.org/10.1126/science.1255484).
- Ioannidis, John P. A., T. D. Stanley, and Hristos Doucouliagos.** 2017. “The Power of Bias in Economics Research.” *Economic Journal* 127 (605): F236–F265. [10.1111/eoj.12461](https://doi.org/10.1111/eoj.12461).
- La Porta, Rafael, Florencio Lopez-de Silanes, Andrei Shleifer, and Robert W. Vishny.** 1998. “Law and Finance.” *Journal of Political Economy* 106 (6): 1113–1155. [10.1086/250042](https://doi.org/10.1086/250042).

A Keyword Taxonomy

Table A1 lists all terms used in Pass 1 of the classification pipeline.

Table A1: In-Scope Classification: Keyword Taxonomy

Category	Keywords
Financial regulation	regulation, regulatory, Dodd-Frank, Basel, Glass-Steagall, capital requirements, leverage ratio, bank capital
Securities law	SEC, disclosure requirement, Reg FD, Sarbanes-Oxley, SOX, insider trading law, short-selling ban, short-sale restriction
Banking policy	FDIC, deposit insurance, lender of last resort, bailout, TARP, stress test, bank resolution, government guarantee
Monetary/fiscal policy	quantitative easing, QE, Fed intervention, government subsidy, fiscal stimulus
Market microstructure rules	circuit breaker, tick size, trading halt, uptick rule, limit-down, limit-up, trading curb
Corporate governance law	board independence requirement, say-on-pay, proxy access, mandatory audit, governance mandate, clawback provision, mandatory disclosure
General intervention signals	government intervention, government policy, public policy, law and finance, legal protection, shareholder rights, creditor rights, investor protection

Ambiguous standalone terms (*regulation*, *regulatory*, *SEC*) require co-occurrence with

at least one additional in-scope keyword before the paper is flagged as a candidate.

B LLM Prompts

B.1 In-Scope Classification Prompt

A finance research paper is ‘in scope’ for a study of publication bias in government intervention research if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Exclusion criteria: (1) purely theoretical papers with no empirical test of a policy; (2) papers that only DESCRIBE a regulation without estimating its effect; (3) papers whose main topic is private firm or market behaviour, with policy as incidental context; (4) papers studying central bank communication WITHOUT a specific policy instrument. Return JSON with exactly three keys: `in_scope` (boolean), `confidence` (‘‘high’’, ‘‘medium’’, or ‘‘low’’), `reason` (one sentence).

B.2 Direction Coding Prompt

The following is the abstract of a finance paper that studies the effect of a government law, regulation, or public policy. Classify the MAIN empirical finding as: +1 if the government intervention produced a POSITIVE or BENEFICIAL outcome (e.g., improved market quality, reduced systemic risk, better investor protection); -1 if it produced a NEGATIVE or HARMFUL outcome (e.g., reduced liquidity,

higher costs, unintended consequences, welfare losses); 0 if the finding is NULL, MIXED, or INCONCLUSIVE. Coding rule: code the sign of the estimated coefficient on the primary intervention variable, NOT the authors' normative framing. Return JSON with: direction (1, -1, or 0), confidence ('high', 'medium', or 'low'), rationale (one sentence).

C Additional Tables

Table A2: Probit Results by Sub-Period

	2000–2007	2008–2015	2016–2024
Positive (+1)	1.431*** (0.469)	0.785** (0.382)	1.352*** (0.305)
Negative (−1)	0.641 (0.792)	1.672*** (0.344)	1.143*** (0.259)
N	45	97	154
Pseudo R^2	0.179	0.207	0.160

Notes: Each column estimates equation (1) on the indicated sub-period without additional controls. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The 2000–2007 column has only 45 observations; coefficients should be interpreted with caution. Period 2008–2015 spans the financial crisis and post-crisis regulatory response (Dodd-Frank enacted July 2010); 2016–2024 covers the post-implementation period.

D Data and Replication

All data and code necessary to reproduce the results of this paper are available at [https://github.com/\[REPOSITORY\]](https://github.com/[REPOSITORY]). The repository contains:

- `requirements.txt` — Python package versions used

- `scripts/01_collect_nber.py` — NBER API data collection
- `scripts/02_collect_journals.py` — Journal data collection
- `scripts/02b_enrich_jfe_abstracts.py` — Elsevier API abstract enrichment for *Journal of Financial Economics*
- `scripts/03_classify_inscope.py` — Two-pass in-scope classification (keyword + LLM)
- `scripts/04_code_direction.py` — LLM direction coding
- `scripts/05_link_nber_to_published.py` — Fuzzy title matching for NBER-to-journal linkage (sample period: 2000–2024)
- `scripts/06_analysis_main.py` — Primary probit regressions and summary statistics
- `scripts/08_robustness.py` — Robustness checks
- `data/` — Final analysis dataset in Parquet format
- `paper/` — L^AT_EX source for this paper